



Short Communication

Increased solubilization of excess sludge does not always result in enhanced anaerobic digestion efficiency



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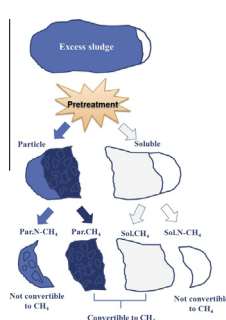
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HIGHLIGHTS

- BMP of whole and soluble fraction of pretreated (alkali + ultrasonication) sludge.
- Decrease of CH₄ conversion of soluble fraction with solubilization increase.
- In case of severe pretreatment (pH 13 + UI 60 min), only 61% was converted to CH₄.
- Soluble but non-biodegradable portion at (pH 13 + UI 60 min) = 25.8%.
- Increased solubilization does not always result in enhanced AD efficiency.

GRAPHICAL ABSTRACT



ARTICLE INFO

Article history:

Received 15 April 2013

Received in revised form 11 June 2013

Accepted 14 June 2013

Available online 22 June 2013

Keywords:

Excess sludge

Anaerobic digestion

Ultrasonic pretreatment

Alkaline pretreatment

Solubilization

ABSTRACT

“Solubilization (SCOD/TCOD, chemical oxygen demand (COD), %)” is generally used as an indicator for the pretreatment effectiveness in anaerobic digestion (AD) of excess sludge. In the present work, ultrasonic (irradiation 5–60 min) and alkaline pretreatments (pH 9–13) were applied to excess sludge individually and in combination. Biological methane potential tests were carried out using the whole part of pretreated sludge and only the soluble fraction. Although solubilization increased with pretreatment intensity increase, methane production was inconsistent with increased solubilization, due to the lower methane yield of soluble fraction. While the soluble fraction obtained under mild pretreatment condition (pH 9 + ultrasonication 5 min) showed 91% of methane yield, it was only 61% in case of severe pretreatment condition (pH 13 + ultrasonication 60 min). 25.8% of the pretreated sludge at (pH 13 + ultrasonication 60 min) consisted of soluble but non-biodegradable portion. These findings demonstrate that increased solubilization does not always result in enhanced AD efficiency.

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1. Introduction

One of the major challenges in wastewater treatment is the management of excess sludge whose treatment cost accounts for up to 60% of the total operating costs of municipal wastewater

treatment plants (WWTPs) (Yan et al., 2013). The amount of sludge produced in Korea has continuously increased with the increasing number of WWTPs, recently reaching 3 million tons of sludge cake per year (MOE, 2011). While the sludge is ultimately disposed by ocean dumping, incineration, or land filling, it is important to minimize its production, not only for economical operation of WWTPs but also for global environmental protection. Anaerobic digestion (AD) is the most traditional and widely used excess sludge treatment method, offering advantages of generation of energy-rich gas in the form of methane (CH₄) along with reduction of sludge

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volume. However, its application is often limited by low overall degradation efficiency (20–50%), which is associated with the hydrolysis step of excess sludge (Kuglarz et al., 2013). The major organic fractions of excess sludge are microbial cells, and their cell walls contain glycan strands, which cause resistance to biodegradation. In order to improve digestion efficiency, numerous disintegration methods such as mechanical, chemical and thermal have hence been applied as pretreatment: these methods disrupt cell walls and readily available intracellular organic materials are released (Appels et al., 2008).

Ultrasonication is recognized as the most powerful method to disintegrate sludge cells. Rapid collapse and expansion of microbubbles generated by cavitation give rise to high-temperature and high-pressure gradients in the liquid phase, leading to rupture of cell walls and membranes (Pilli et al., 2011). Cell disintegration of 100% can be achieved at high power levels, but power consumption then becomes a serious economic burden (Weemaes and Verstraete, 1998). Recently, ultrasonication has often been combined with alkaline pretreatment (Kim et al., 2010; Li et al., 2010). When ultrasonication was applied to the alkali-pretreated sludge at pH 9–11, a high disintegration degree of sludge was obtained at a low energy level of ultrasonication and a synergistic disintegration effect was observed.

Solubilization, defined as the ratio of soluble organics (usually expressed as “chemical oxygen demand (COD)”) to total organics, is generally used as an indicator for the pretreatment effectiveness of excess sludge (Appels et al., 2008; Bougrier et al., 2005). However, according to our findings, increased solubilization does not always result in enhanced AD efficiency, particularly, when a high strength level of pretreatment is applied. Wilson and Novak (2009) reported that temperature higher than 180 °C led to the production of recalcitrant soluble organics or toxic/inhibitory intermediates, thereby reducing the biodegradability of the sludge. According to Park et al. (2012), although combined alkaline and ultrasonic pretreatment increased the initial rate of CH₄ production by increasing solubilization, it had a negligible effect on the ultimate CH₄ yield. Bougrier et al. (2006) found that, in spite of the same solubilization, sludge pretreated by ultrasonication showed higher biodegradability than those pretreated by ozonation and thermal treatment. These findings have following important implications: (1) some part of the soluble fraction generated by pretreatment is not converted to CH₄; (2) some part of particle fraction weakened by pretreatment but not solubilized is converted to biogas; and, finally, (3) increased solubilization cannot be directly linked to improved digestion efficiency.

In the present work, we investigated the relationship between increased solubilization and AD efficiency at various pretreatment conditions. Ultrasonic and alkaline pretreatments were applied to excess sludge individually and in combination. Solubilization at various pretreatment conditions was measured, and biological methane potential (BMP) tests were carried out using not only the whole part of pretreated sludge but also the soluble fraction. The sludge was fractionated into four parts to ascertain the major constituent of total CH₄ produced and attain a deep understanding on the efficiency of pretreatment.

2. Methods

2.1. Excess sludge and inoculum

The excess sludge used in this experiment was taken from the sludge thickener line of Daejeon WWTP. The total COD concentration of excess sludge was 19.6 g/L while 6% of the total sludge were soluble. Total nitrogen (TN) concentration was 1.2 g/L and pH was 7. From the same WWTP, the inoculum, flocculated sludge, for the

BMP test was obtained from an anaerobic digester. The pH, alkalinity, and volatile suspended solids (VSS) concentration of the sludge were 7.5, 2.83 g CaCO₃/L, and 5.3 g/L, respectively.

2.2. Disintegration methods

Ultrasonic treatment was performed using an ultrasonicator with a frequency of 20 kHz (VCX-750, Sonics and Materials, USA). Ultrasonic treatment (750 W, amplitude = 40%) was applied to 100 mL of excess sludge for 5, 10, 20, 40, and 60 min. In the alkaline pretreatment, a 6 mol/L KOH solution was added to 300 mL of excess sludge in order to reach pH values of 9, 10, 11, 12, and 13. The samples were mixed for 1 h using a magnetic stirrer at 200 rpm. The expected pH levels were achieved instantaneously, but dropped slightly with time. Therefore, the injection of alkaline solution was continued throughout the hour. In the combined pretreatment, alkali-pretreated sludge at pH 9, 11, and 13 for 1 h was ultrasonicated for 5, 20, and 60 min, totally 9 combinations. After pretreatment, the sludge was centrifuged (10,000 rpm for 20 min) and filtered through a 0.45 µm pore size membrane filter to obtain the soluble fraction.

2.3. BMP test and analysis

BMP tests were carried out using a 250 mL serum bottle (effective volume of 100 mL). After placing 69 mL of inoculum, 30 mL of pretreated excess sludge (whole part or soluble part), and 1 mL of concentrated trace element solution into the serum bottle, N₂ gas was purged for 5 min to establish an anaerobic condition. A trace element solution was made according to Kim et al. (2006). The initial pH was set to 7.5 ± 0.1 using 3 mol/L KOH and 3 mol/L HCl. The bottles were incubated at 35 °C using a water bath with agitation at 150 rpm using a shaking incubator. The tests were carried out in triplicate and the results were averaged.

The volume of produced biogas was measured by the displacement of a glass syringe piston, and was converted to a value at standard temperature and pressure (STP, 0 °C and 1 bar) conditions. Biogas at the head space in the reactors was collected using a gas-tight micro syringe, and the biogas composition was analyzed by a gas chromatograph (Gow-Mac Series 580, USA) equipped with a thermal conductivity detector and a stainless steel column. The temperatures of the injector, column, and detector were kept at 50, 80, and 90 °C, respectively. The carrier gas was N₂ and the flow rate was 30 mL/min. The concentrations of COD, TN, alkalinity, and VSS were measured according to standard methods in Part 5220, Part 4500, Part 2320, and Part 2540, respectively (APHA, 2005).

3. Results and discussion

3.1. Ultrasonication

As listed in Table 1, the increase of ultrasonication strength resulted in the increase of both solubilization and CH₄ yield of excess sludge, indicating that the ultrasonication applied in this experimental range was effective for enhancing AD efficiency. As irradiation time increased to 60 min, solubilization and CH₄ yield continuously increased from 6% to 33% and 25.7% to 43.2%, respectively. CH₄ yield was obtained by comparing the actual amount of CH₄ produced to the theoretical maximum (1 g COD = 350 mL CH₄) (Kim et al., 2010).

Solubilization of raw (untreated) sludge was 6% and 75.4% of the soluble fraction was converted to CH₄, which contradicts the general understanding that most of the soluble fraction can be converted to CH₄ by AD. With 5 min of ultrasonication, the CH₄ yield of

Table 1Effect of pretreatment on the solubilization and CH₄ production from excess sludge.

Pretreatment	^a Solubilization (%)		CH ₄ production (mL CH ₄ /100 mL bottle)		^b CH ₄ yield of whole sludge (A, %)	^b CH ₄ yield of soluble fraction (B, %)	^c CH ₄ yield of particle fraction (C, %)
			Whole sludge	Soluble fraction			
Control (untreated)		6	53 ± 2	9 ± 1	25.7	75.4	22.5
Ultrasonication duration (min)	5	12	74 ± 3	23 ± 2	35.9	94.0	27.9
	10	19	79 ± 5	35 ± 2	38.3	89.1	26.3
	20	24	80 ± 6	42 ± 2	38.8	86.2	23.9
	40	29	88 ± 7	50 ± 5	42.7	83.0	26.2
	60	33	89 ± 5	51 ± 4	43.2	75.3	27.2
Alkaline pretreatment (pH)	9	11	75 ± 2	22 ± 2	36.4	96.9	28.9
	10	15	78 ± 3	28 ± 1	37.9	88.1	28.9
	11	20	74 ± 2	36 ± 3	35.9	85.0	23.4
	12	26	71 ± 4	41 ± 2	34.5	75.9	20.0
	13	28	65 ± 3	39 ± 3	31.6	67.0	17.8
	9–5	27	86 ± 5	51 ± 3	42.0	91.1	23.7
Combined pretreatment (pH-irradiation time)	9–20	39	87 ± 4	60 ± 4	42.1	74.1	21.4
	9–60	48	91 ± 9	69 ± 5	44.4	70.6	20.1
	11–5	36	95 ± 4	56 ± 3	46.2	75.6	29.5
	11–20	50	97 ± 9	69 ± 6	47.3	67.3	27.2
	11–60	59	102 ± 6	74 ± 5	49.4	61.1	32.4
	13–5	42	94 ± 7	56 ± 2	45.8	65.3	31.7
	13–20	53	106 ± 7	70 ± 3	51.7	64.2	37.5
	13–60	65	103 ± 5	82 ± 5	50.2	60.5	30.9

^a Solubilization = (SCOD/TCOD) × 100.^b A, B = (Actual amount of CH₄ produced/theoretical maximum amount of CH₄ produced) × 100.^c C = (A – (Solubilization × B))/(100 – Solubilization) × 100.

the soluble fraction increased to 94%, but it gradually decreased with irradiation time increase. This is clear evidence that some soluble matter that is not convertible to CH₄ could be generated by pretreatment.

To know the major constituent of total CH₄ produced, the pretreated sludge was fractionated into four parts on COD basis: soluble and convertible to CH₄ (Sol.CH₄), soluble but not convertible to CH₄ (Sol.N-CH₄), particle but convertible to CH₄ (Par.CH₄), and particle and not convertible to CH₄ (Par.N-CH₄) (Fig. 1), which were calculated according to Eqs. (1)–(4).

$$\text{Sol.CH}_4 = (\text{Solubilization} \times \text{CH}_4 \text{ yield of soluble fraction}) \times 100 \quad (1)$$

$$\text{Sol.N-CH}_4 = \text{Solubilization} - \text{Sol.CH}_4 \quad (2)$$

$$\text{Par.CH}_4 = \text{CH}_4 \text{ yield of whole sludge} - \text{Sol.CH}_4 \quad (3)$$

$$\text{Par.N-CH}_4 = 1 - \text{Solubilization} - \text{Par.CH}_4 \quad (4)$$

It was found that the raw sludge consisted of 4.5% Sol.CH₄, 1.5% Sol.N-CH₄, 21.2% Par.CH₄, and 72.8% Par.N-CH₄, indicating that most of the produced CH₄ originated from the particle fraction (Fig. 2). As the ultrasonication strength increased, the portion of Sol.CH₄ increased, which resulted in an increased total CH₄ yield. However, the enhanced AD efficiency cannot be entirely ascribed to the increased solubilization, since the Sol.N-CH₄ portion also increased with irradiation time increase and more than 40% of the total CH₄ produced still originated from the particle fraction even at 60 min irradiation. It was interesting to see that the CH₄ yield of particle fraction (Table 1) was slightly increased by ultrasonication. This is clear evidence that some parts of the particle fraction weakened by pretreatment but not solubilized were converted to CH₄.

3.2. Alkaline pretreatment

Unlike in ultrasonication, although solubilization increased with alkaline pH increase (pH 9–13), the CH₄ yield was maximized at pH 10. This clearly indicates that conversion of excess sludge to

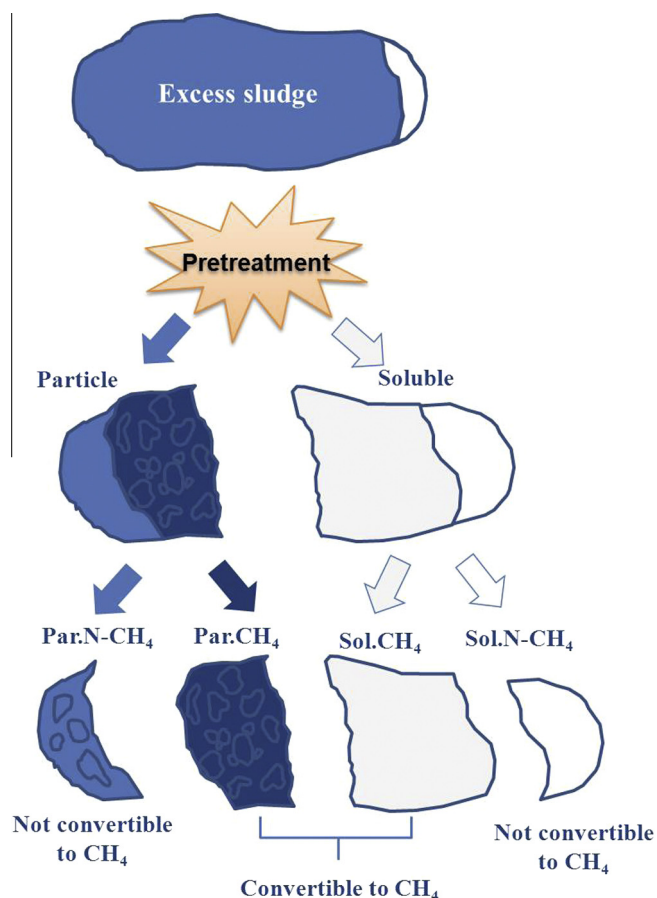


Fig. 1. Division of pretreated excess sludge into four parts (Par.N-CH₄: particle and not convertible to CH₄, Par.CH₄: particle but convertible to CH₄, Sol.CH₄: soluble and convertible to CH₄, Sol.N-CH₄: soluble but not convertible to CH₄).

soluble matter by pretreatment is not always beneficial for AD. Consistent with our study, Li et al. (2012) observed a discrepancy

between the solubilization increase and biogas production increase, and attributed this to the increase of initial pH increase and salinity by NaOH addition. However, in our study, alkali-treated sludge was neutralized after 1 h of alkaline pretreatment, and KOH was used as an alkaline solution. Compared to sodium, potassium has negligible effect on AD (Kim et al., 2009).

At the same solubilization degree, lower CH_4 yields were obtained with alkaline pretreatment than with ultrasonication, which could be attributed to the low CH_4 yield of both the soluble and particle fractions. At similar solubilization degree (29% at ultrasonication for 40 min and 28% at pH 13), 19% and 35% lower CH_4 yields of soluble fraction and particle fraction, respectively, were achieved with alkaline pretreatment. The low CH_4 yield of soluble fraction indicates the production of refractory soluble organics by pretreatment. Unlike in ultrasonication, the CH_4 yield of the particle fraction decreased with alkali pH increase, resulting in a slight variation of Par.N- CH_4 (Fig. 1). This indicated that the remaining particle fraction obtained after ultrasonication was more biodegradable than that obtained with alkaline pretreatment. Structural analysis on the particle fraction using a scanning electron microscope (SEM) and transmission electron microscope (TEM) is required to support this conclusion.

3.3. Combined (alkaline + ultrasonic) pretreatment

In the combined pretreatment, when ultrasonication was irradiated for 60 min to the alkali-pretreated sludge at pH 13 (pH 13-UI

60 min), solubilization increased to 65%. However, only 61% of the soluble fraction was converted to CH_4 ; this may be due to the production of refractory soluble organics under the severe pretreatment condition. Meanwhile, 91% of the soluble fraction was converted to CH_4 in the mild pretreatment condition of (pH 9-UI 5 min). Because of this big difference in the CH_4 yield of the soluble fraction between the two cases, there was only a 20% increase in AD efficiency (CH_4 yield of whole sludge) in spite of 2.4 times higher solubilization being achieved with (pH 13-UI 60 min). It was also notable that, compared to individual pretreatment, a much larger portion of excess sludge was converted to Sol.N- CH_4 , especially under severe pretreatment conditions. The Sol.N- CH_4 portion in (pH 13-UI 60 min) and (pH 9-UI 5 min) were 25.8% and 2.4%, respectively.

This study has significant meaning in that it reports first results on how much of the soluble fraction in pretreated excess sludge is converted to CH_4 and clearly demonstrates that increased solubilization does not always result in enhanced AD efficiency. Although, there have been some studies to date on the compounds comprising Sol.N- CH_4 in thermal pretreatment, little effort has been made when other disintegration methods were applied (Bougrier et al., 2007; Wilson and Novak, 2009). At high temperature, low molecular weight intermediates, such as carbohydrates and amino compounds, are polymerized with the formation of compounds with a molecular weight of 50–70 kDa, presenting low biodegradability. Future work would focus on analyzing not only the particle fraction before and after AD such as proteins, polysaccharides, lipids, but also the soluble fraction in terms of molecular weight and excitation–emission matrix.

4. Conclusion

From the BMP tests using the whole pretreated sludge and only the soluble fraction, the sludge was fractionated into four parts: Sol. CH_4 , Sol.N- CH_4 , Par. CH_4 , and Par.N- CH_4 . As the pretreatment intensity increased, the CH_4 yield of the soluble fraction decreased, resulting in an increase of Sol.N- CH_4 portion. Sol.N- CH_4 portion in (pH 13-UI 60 min) and (pH 9-UI 5 min) were 25.8% and 2.4%, respectively, resulting in a little difference in AD efficiency. Even at severe pretreatment conditions, a considerable amount of CH_4 originated from the particle fraction. These findings demonstrate that increased solubilization does not always result in enhanced AD efficiency.

Acknowledgements

This work was supported by the Intelligent Synthetic Biology Center of Global Frontier Project funded by the Ministry of Education, Science and Technology (2011-0031955) and was conducted under the framework of Research and Development Program of the Korea Institute of Energy Research (KIER) (B3-2423).

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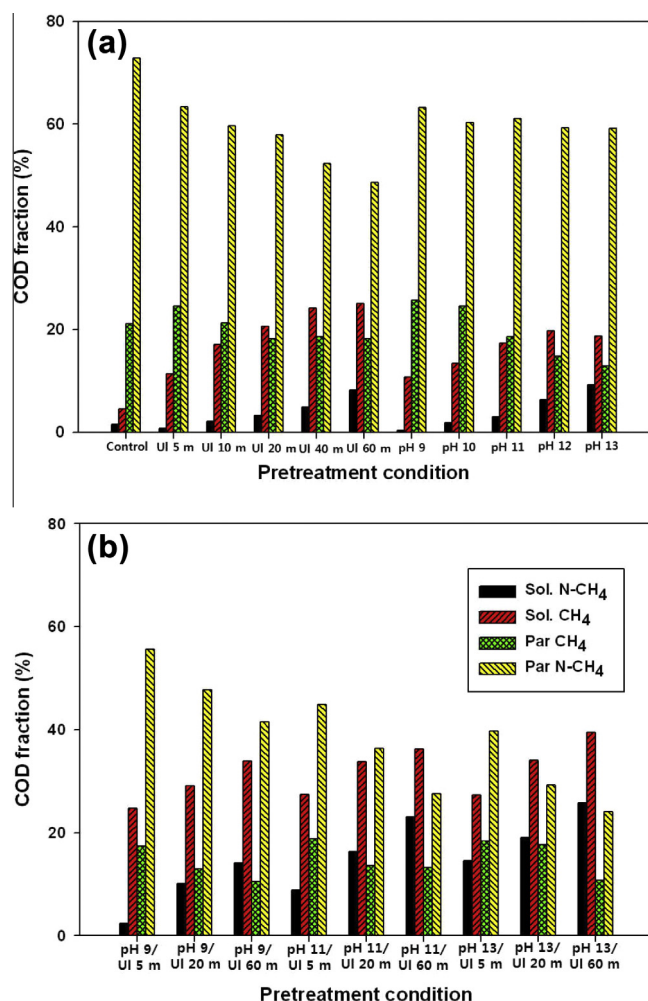


Fig. 2. Division of excess sludge in the (a) individual and (b) combined pretreatment.

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